Guarding Self-Evolving AI with Proofs

llcore research findings 2026 — verified neural evolution on CPU

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github.com/furuse-kazufumi/llcore (paper draft + all experiment code/data public)

Problem: on what grounds may a self-evolving AI admit its own changes?

AI now evolves its own core (memory, dynamics) — how do we reject dangerous changes?
Mainstream (DGM / SEAL line) = empirical gates: benchmark scores, past failures
Structural weakness: (1) learns by dying — pays the failure cost (2) Goodhart-able — metrics can be gamed
llcore's question: if we narrow the property, can a mathematical proof be the gate?

Approach: a sound contraction-certifier gate (fail-closed)

Certified property = contraction ($\rho < 1$): internal state cannot diverge
Three sound certifiers: closed-form ∞-norm ($O(n^2)$) → vertex 2-norm → Lyapunov SDP
Fail-closed: unprovable changes are rejected. Proofs are theorem-derived = non-fabricable
Gating inside the evolution loop, before evaluation — 17-19x cheaper than post-hoc certification

All proofs & experiments pre-registered before results (research/paper/PAPER_DRAFT.md)

Testbed: a lethal environment x three ways experience becomes memory

The environment steps the recurrence gain κ by 2x → previously-safe genes diverge = death
ENDO (endogenous foresight): a sound self-held certificate rejects lethal changes before evaluation
REVIVE (certificate-preserving repair): dies, but is revived with its memory gene intact
OBSERVE (observational): empirically avoids the neighborhood of others' observed deaths
n=20 seeds, pre-registered, steady-state after warm-up (all negatives disclosed)

Finding 1: proofs know before dying — zero steady-state deaths

Lethal evaluations (steady state, linear substrate):
無防備 27.2 / 観察学習 17.3 / 固定ゲート 10.2 / 証明 (ENDO) 0.0
Observational learning works — but deaths remain. Learning-by-dying is a structural cost
Robustness: the proof's advantage holds in 12/12 pre-registered configurations
And the proof gate does not strangle memory capability (worst fitness gap 0.7%)

$p < 0.001$ (paired sign-flip, n=20). All numbers in public results_viability_*.json

Finding 2: without revival, experience does not become memory

Death removes the individual — and erases the experience its genes accumulated
REVIVE dies about as often as no-gate, yet preserves significantly more population memory
集団平均適応度: 無防備 0.649 → 復活あり 0.708 (Δ=+0.060, p=0.0011)
Repair preserves the memory channel while safing the dynamics — death leaves a scar, not a void

Finding 3: one proof beats stacked empirical mechanisms

All 8 mechanism combinations compared under identical conditions (2^3 factorial):

- 証明 (E) を含む 4 組み合わせ: 全て死 0.0
- 証明なしの最良 = 復活+観察 (RO): 6.2 — 単独 (18.8 / 17.3) より良いが 0 には届かない

With the proof present, empirical mechanisms add nothing (interaction p=0.0003/0.0045)

Empirical mechanisms do compose well — but only in a world without the proof

Finding 4 (limits): a proof is only sound while its premises hold

If the verifier under-senses the environment by 40%, the certificate admits a lethal band

A trust controller using only observable evidence (admitted-then-died) can switch proof <-> empirical

But hedging pays only where premise violations are costly — a pre-registered hypothesis failed on one substrate and we report it

Open question: who certifies the certifier's premises? (premise monitoring)

PoC scale (scalar genes, n=20) — not yet shown at real-LLM scale. No overclaim

Map: how this differs from the nearest work (primary sources verified)

SGM (Statistical Goedel Machine): statistical certificate over discrete task scores — probabilistic, not fabrication-proof

SEVerA: sound Dafny-verified logical I/O contracts — discrete artifacts, not continuous internal dynamics

PAC limits of self-improvement (Wang et al.): generalization bounds — not stability/contraction

The open quadrant — continuous memory dynamics, gated in-loop by a sound contraction proof — is what llcore occupies at PoC scale

All three checked against primary sources with adversarial review. Full contrasts in the paper's related-work section

Using this work

These slides: CC BY 4.0 — free for corporate training, tech scouting, product evaluation with attribution
Code / paper draft: Apache-2.0 + Commercial dual license (github.com/furuse-kazufumi/llcore)
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